EARLY TREND DETECTION USING CLOUD-BASED ANALYTICS & POWER BI

A Project Report
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In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled **"Early Trend Detection using Cloud-Based Analytics and Power BI"** is a record of Bonafide work of team members **Gangasani Lavanya – 2320090004, Pendyala Abhinav – 2320030215, and Thigulla Adithya Kiran Goud – 2320030498**, submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

This is to certify that the project-based report entitled **"Early Trend Detection using Cloud-Based Analytics and Power BI"** is a Bonafide work carried out and submitted by **Gangasani Lavanya – 2320090004, Pendyala Abhinav – 2320030215, and Thigulla Adithya Kiran Goud – 2320030498,** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, at **K L (Deemed to be University)**, during the academic year **2025–2026**.

Signature of the Supervisor

Signature of the HOD

Signature of the External Examiner

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ABSTRACT

In the modern digital era, information spreads rapidly across search engines, news platforms, and online communities. Detecting such trends at an early stage has become highly important for businesses, media organizations, sports analytics platforms, and decision-making systems.

This project, titled **"Early Trend Detection using Cloud-Based Analytics & Power BI"**, focuses on designing and implementing a real-time cloud-powered analytics system that continuously detects emerging and viral trends from live internet search activity.

The system uses real-time trending topic data obtained from **Google Trends RSS feed**, which acts as the live data source. This data is continuously fetched using a **Python-based** ingestion script and streamed into **Microsoft Azure** cloud services.

Azure Event Hub is used as the real-time event ingestion service to capture and manage continuous data streams. The incoming events are then processed using **Azure Stream Analytics**, where live aggregation, trend scoring, engagement calculation, and trend classification logic are applied.

The processed results are visualized using **Power BI** dashboards, where multiple visual components such as **line charts, KPI cards, and trend status tables** are used to present live analytics insights.

The system calculates key parameters such as mention count, average engagement, peak engagement, and trend status. Based on **engagement thresholds**, each topic is classified into categories such as **Viral, Emerging, and Normal.**

The project demonstrates how cloud-based streaming analytics can be effectively used for real-time trend monitoring and early spike detection.

The results clearly show that the system is capable of detecting live trends, tracking their intensity over time, and presenting meaningful insights through interactive dashboards.

This project highlights the practical application of **Microsoft Azure** cloud services and business intelligence tools in building scalable, real-time analytics systems.

INTRODUCTION

In today's highly connected digital environment, information spreads faster than ever before. Search engines, news platforms, sports portals, and online media channels generate a massive volume of continuously changing data every second.

Events such as breaking news, cricket match moments, movie releases, political announcements, celebrity updates, and viral internet discussions can become trending topics within minutes.

For organizations operating in domains such as media, sports analytics, marketing, and business intelligence, identifying these emerging trends at an early stage is extremely important.

Traditional reporting systems are mostly static and retrospective in nature.
They provide insights based on historical data and usually answer the question:

"What has already happened?"

However, modern organizations require systems that answer:

"What is rising right now?"
"What is likely to become the next major trend?"

This need for proactive insight forms the foundation of this project.

The project titled **"Early Trend Detection using Cloud-Based Analytics & Power BI"** focuses on building a real-time cloud-based trend intelligence system.

The primary objective of the system is to capture live trend signals from internet search activity, process them in real time using cloud infrastructure, and visualize them through interactive dashboards.

This project uses **Google Trends** live data as the source of trending topics.
These topics are then streamed into **Microsoft Azure** cloud services for
continuous event processing.

Azure Event Hub is used for real-time data ingestion, **Azure Stream Analytics** is used for live aggregation and trend scoring, and **Power BI** is used for dashboard visualization and business intelligence reporting.

Problem Description and Objectives

Problem Description

In the modern digital ecosystem, trends emerge and evolve dynamically. A topic that is unknown at one moment may become viral within minutes. This rapid change creates a challenge for organizations that rely on timely decision-making. Traditional analytics tools are not designed to process continuous live streams of incoming events. They often depend on batch processing and delayed reporting mechanisms, which makes early trend identification difficult. As the volume and velocity of data increase, manual analysis becomes impractical and error-prone. This creates the need for a cloud-based scalable architecture capable of processing real-time data streams.

Objectives

The main objectives of this project are:

1. To collect live trending data from **Google Trends** in real time.

2. To stream incoming events using **Microsoft Azure Event Hub.**

3. To process and aggregate live event streams using **Azure Stream Analytics**.

4. To calculate key metrics such as:
   - mention count
   - average engagement
   - peak engagement

5. To classify live trends into categories such as:
   - Viral
   - Emerging
   - Normal

6. To visualize trend movement using **Power BI** dashboards.

7. To enable early detection of rising trends before they fully peak.

Through this project, the importance of cloud-based real-time analytics and business intelligence systems is clearly demonstrated.

LITERATURE SURVEY

The rapid growth of digital platforms, search engines, and online content distribution systems has led to an exponential increase in continuously changing data streams.

Modern organizations rely heavily on trend intelligence to make fast and effective decisions.

Real-time analytics has therefore become a major area of research in cloud computing, business intelligence, and big data systems.

Several researchers have explored the use of cloud-based streaming systems for continuous event monitoring and real-time insight generation.

Traditional data processing techniques are often batch-oriented and are not suitable for handling live streams of high-velocity data.

This limitation has led to the widespread adoption of cloud platforms such as **Microsoft Azure, Amazon Web Services, and Google Cloud Platform.**

These cloud systems provide scalable infrastructure for real-time event ingestion, processing, and visualization.

Cloud computing has significantly transformed the way modern organizations collect and process data.

According to *Armbrust et al. (2010)*, cloud platforms provide scalable and elastic computing resources that allow systems to process large volumes of incoming data without investing in dedicated physical infrastructure.

This is highly beneficial in applications where data changes continuously, such as news analytics, social media monitoring, and search trend analysis.

Streaming data systems have emerged as a major research domain due to the need for processing live events in near real time.

Researchers have shown that event-driven architectures are highly effective for handling continuous data flows.

Cloud-Based Streaming Analytics

Cloud-based streaming analytics has become an essential component of modern business intelligence systems. **Microsoft Azure** provides specialized services such as **Azure Event Hub** and
Azure Stream Analytics, which are designed specifically for high-throughput stream processing.

Azure Event Hub is widely used in enterprise environments for ingesting millions of events per second. Research studies have shown that such event ingestion systems significantly improve scalability and fault tolerance compared to traditional systems.

Azure Stream Analytics enables real-time **SQL-like** processing on incoming streams. This allows live aggregation, filtering, windowing, and rule-based analytics. According to **Microsoft's** cloud analytics documentation, stream analytics services are particularly effective for anomaly detection, live dashboards, and event-based intelligence systems.

Business intelligence tools such as **Power BI** have also been extensively studied in the field of interactive data visualization. *Few (2009)* emphasized the importance of converting complex data into meaningful visual insights.

Interactive dashboards improve decision-making by allowing analysts to observe patterns, anomalies, and rising signals in real time. **Power BI** is widely used for **KPI monitoring**, **trend visualization**, and **interactive business dashboards**.

Trend Detection and Decision Support

Trend detection has become an important research area in data science and business intelligence. Several studies have shown that early detection of rising patterns can significantly improve strategic decision-making.

Trend intelligence systems are commonly used in:
- media monitoring
- financial analytics
- sports analytics
- marketing intelligence
- public sentiment tracking

Researchers have proposed multiple approaches for trend detection, including:
- moving averages
- spike detection
- threshold-based analytics
- time-series forecasting
- anomaly detection

Among these, **threshold-based real-time classification** remains one of the most interpretable and effective approaches for live systems. This project builds upon these concepts by integrating **real-time data collection, cloud event streaming,** and **dashboard-based visualization**.

The literature clearly demonstrates that **cloud-powered analytics** platforms are highly suitable for real-time trend monitoring applications. This project extends these research concepts into a practical implementation using **Microsoft Azure** and **Power BI.**

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Specifications

The **"Early Trend Detection using Cloud-Based Analytics & Power BI"** project does not require high-end local hardware because the major processing operations are performed in the cloud using **Microsoft Azure**.

However, a standard computing system is required to access cloud services, execute Python scripts, and visualize dashboards smoothly.

Processor

A system with **Intel Core i3** processor or equivalent is sufficient for running the project. For better performance and multitasking, **Intel Core i5** or above is recommended.

Since the majority of data processing is performed in Azure cloud services, the local processor is mainly used for:
- running the **Python** data ingestion script
- accessing **Azure Portal**
- working with **Power BI**
- report documentation

RAM

A minimum of **4 GB RAM** is required. For smooth multitasking and better browser performance, **8 GB RAM** or above is recommended.

This is especially useful while simultaneously running:
- **Python** scripts
- browser tabs
- **Azure dashboard**
- **Power BI reports**

Storage

The project requires very limited local storage because the main data processing and storage operations are cloud-based.

A minimum of **500 MB free disk space** is sufficient for:
- **Python** libraries
- temporary files
- project scripts
- screenshots and reports

Hardware Specifications

Graphics

No dedicated graphics card is required. Integrated graphics available in modern processors are sufficient. Since the project focuses on analytics and dashboard visualization, no **GPU-intensive rendering** is involved.

Operating System

The project is compatible with:

- Windows
- Linux
- macOS

The project has been primarily implemented and tested on **Windows** environment. Because the system is cloud-based, the operating system does not significantly affect execution.

Software Specifications

The software stack used in this project consists of cloud services, programming tools, and visualization platforms.

Cloud Platform

Microsoft Azure

The following Azure services are used:

1. Azure Event Hub
Used for real-time event ingestion and stream management.

2. Azure Stream Analytics
Used for live event processing, aggregation, and trend detection.

3. Azure Portal
Used for service management and monitoring.

Visualization Tool

Power BI

Power BI is used for:
- line charts
- KPI cards
- live dashboards
- trend tables
- report visualization

Software Specifications

Programming Language

Python 3.10+

Python is used for:
- data fetching
- RSS parsing
- event streaming
- JSON payload creation

Libraries Used

1. feedparser
Used for fetching live Google Trends RSS feed.

2. azure-eventhub
Used for sending live events into Azure Event Hub.

3. json
Used for structured payload formatting.

4. datetime
Used for timestamp generation.

5. time
Used for controlled streaming intervals.

Development Environment

The development and execution environment consists of:

- Visual Studio Code
- Command Prompt / Terminal
- Azure Portal
- Power BI Service

Browser Requirements

Google Chrome or **Microsoft Edge** is recommended for smooth access to **Azure Portal** and **Power BI**.

The above hardware and software configuration is sufficient for efficient execution of the project. Since the major computational tasks are handled in the cloud, the system remains lightweight on local resources while providing scalable real-time analytics.

IMPLEMENTATION

The implementation of the **"Early Trend Detection using Cloud-Based Analytics & Power BI"** project focuses on building a scalable real-time analytics pipeline using cloud technologies.

The system is designed to continuously collect live trending topics, stream them into the cloud, process them in real time, and visualize them through interactive dashboards.

The implementation is divided into multiple stages:

1. Live Data Collection
2. Event Streaming
3. Real-Time Processing
4. Dashboard Visualization
5. Trend Classification

Each stage plays a crucial role in ensuring continuous live trend monitoring and early spike detection.

System Architecture

The complete implementation architecture is as follows:

Google Trends RSS Feed
↓
Python Ingestion Script
↓
Azure Event Hub
↓
Azure Stream Analytics
↓
Power BI Dashboard

The system follows an event-driven architecture.

This architecture ensures that incoming trend signals are captured, processed, and visualized without significant delay.

1. Live Data Collection and Ingestion

The live data source used in this project is **Google Trends RSS feed.**

This feed provides continuously updated trending search topics based on live internet search activity. A **Python-based** ingestion script is developed to fetch the **RSS feed** at regular time intervals.

The Python script performs the following steps:

1. Connects to **Google Trends RSS feed**
2. Parses the incoming **XML feed**
3. Extracts trend titles
4. Generates structured **JSON payloads**
5. Adds timestamp
6. Calculates engagement score
7. Sends event into **Azure Event Hub**

Example event structure:

```
{
   "timestamp": "2026-04-07T10:30:00",
   "keyword": "IPL 2026",
   "mentions": 1,
   "engagement": 900
}
```

The script runs continuously at fixed intervals to simulate real-time live event streaming.

2. Cloud Processing and Stream Analytics

Azure Event Hub is used as the real-time event ingestion layer. Its role is to capture and manage the continuous incoming stream of trend events. Each event received from **Python** is pushed into the **Event Hub**, which acts as a scalable event queue. **Azure Stream Analytics** is used as the real-time processing engine. The incoming events are continuously read and processed using SQL-based streaming queries.

The following metrics are calculated:

- mention_count
- avg_engagement
- peak_engagement
- trend_status

The system uses tumbling window aggregation for every 5 minutes. **Trend classification** logic is applied using threshold rules. The logic is as follows:

avg_engagement >= 700 → **Viral**
avg_engagement >= 400 → **Emerging**
below 400 → **Normal**

This helps in early classification of rising topics.

3. Dashboard Visualization

Power BI is used as the visualization layer of the project. The processed stream output from **Azure Stream Analytics** is directly connected to **Power BI** dataset.

The dashboard consists of the following visual components:

1. Line Chart

Displays keyword-wise trend movement over time.

2. KPI Cards

Shows:
- mention count
- average engagement
- peak engagement

3. Trend Status Table

Displays:
- keyword
- trend status
- engagement score
- timestamp

This dashboard provides real-time visibility into rising trends. The line chart clearly shows sudden spikes, making it easy to identify viral topics early.

The dashboard is updated continuously as new events arrive.

EXPEREMENTATION AND CODE:

The project was implemented using Python, Microsoft Azure cloud services, and Power BI.

The codebase is divided into the following major modules:

1. Data Collection Module
Responsible for fetching live trending topics from Google Trends RSS feed.

2. Event Streaming Module
Responsible for converting incoming data into JSON events and sending them into Azure Event Hub.

3. Cloud Processing Module
Azure Stream Analytics query logic for real-time aggregation and classification.

4. Dashboard Visualization Module
Power BI dataset and report configuration.

Python Code Structure

The Python script performs the following operations:

- fetches live RSS feed
- parses trend titles
- creates event payload
- attaches timestamp
- calculates engagement score
- streams events into Azure

The **main libraries** used are:

- feedparser
- azure-eventhub
- datetime
- json
- time

The script is designed to continuously execute at fixed intervals, thereby simulating real-time trend streaming.

Experimentation and Testing

The system was tested in multiple stages to validate the correctness of each component.

Step 1: Live Feed Testing

The Google Trends RSS feed was first tested independently to ensure that live trending topics were being fetched correctly. The output successfully displayed live keywords.

Step 2: Event Hub Validation

The Python script was connected to Azure Event Hub. The incoming events were verified using Azure Data Explorer. This confirmed that live trend events were reaching the cloud successfully.

Step 3: Stream Analytics Validation

The Stream Analytics job was started and connected to Event Hub. The input preview confirmed that events were being received.

The **SQL query logic was tested for:**

- mention count
- engagement aggregation
- threshold classification

Step 4: Power BI Validation

The output dataset was connected to Power BI. The line chart, KPI cards, and trend tables were validated. The dashboard successfully updated with live data.

Performance Analysis

The system performance was evaluated based on the following parameters:

1. Processing Speed

The cloud-based architecture enabled near real-time updates. The average dashboard refresh delay was minimal.

2. Scalability

The Azure-based architecture is capable of handling continuous high-volume event streams.

3. Reliability

The system consistently received live events without data loss.

4. Dashboard Responsiveness

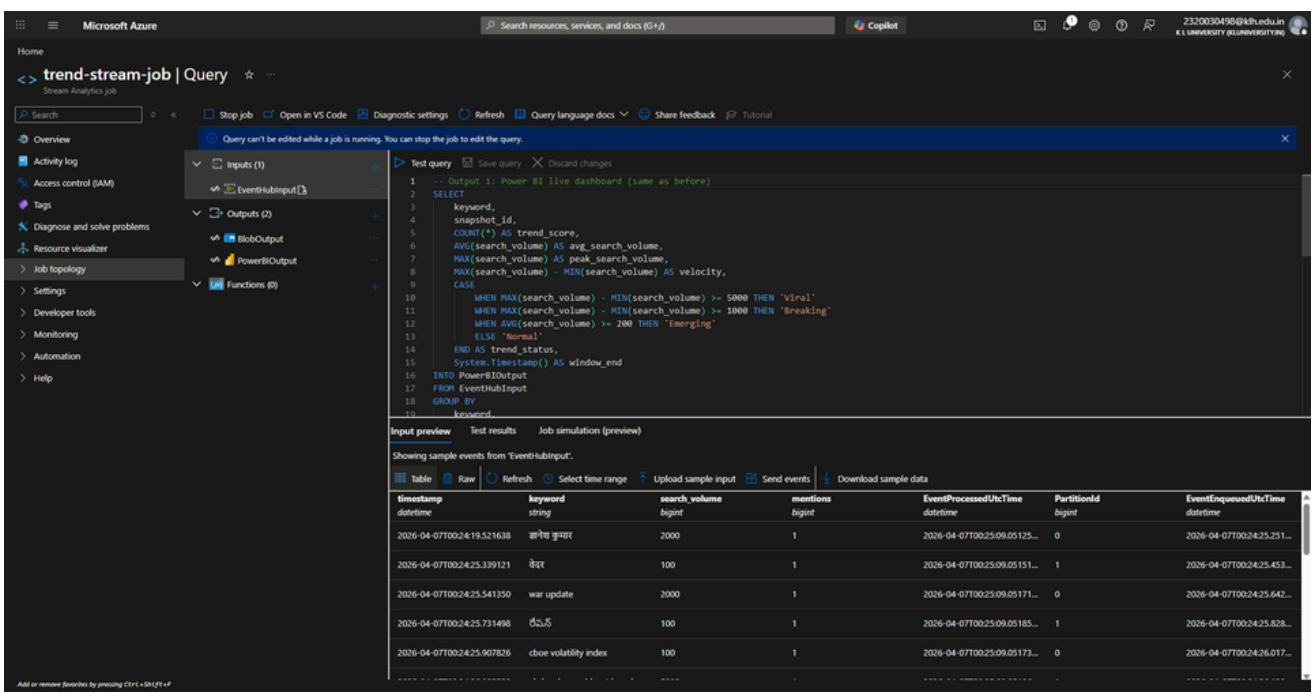
Power BI visuals updated continuously and accurately reflected incoming trend movement.

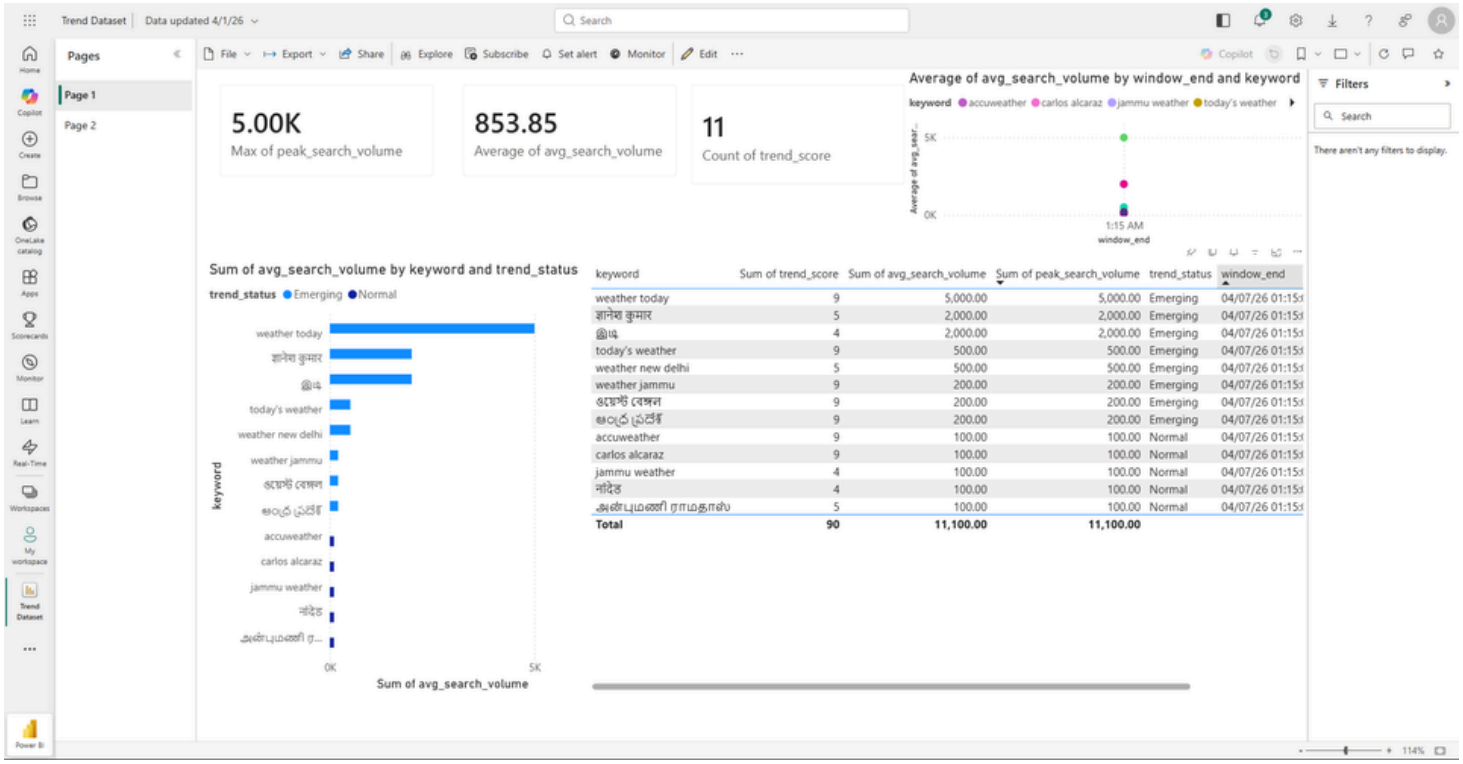
Conclusion of Experimentation

The experimentation phase clearly demonstrates that the proposed cloud-based trend detection system is stable, scalable, and effective for live trend monitoring.

```python
from azure.eventhub import EventHubProducerClient, EventData
from datetime import datetime
import json
import time

CONNECTION_STR = "Endpoint=sb://trendhubadi01.servicebus.windows.net/;SharedAccessKeyName=RootManageSharedAccessKey;SharedAccessKey=GuzvWGE
EVENT_HUB_NAME = "trend-events"
RSS_URL = "https://trends.google.com/trending/rss?geo=IN"

producer = EventHubProducerClient.from_connection_string(
    conn_str=CONNECTION_STR,
    eventhub_name=EVENT_HUB_NAME
)

while True:
    feed = feedparser.parse(RSS_URL)
    print("\n🔥 LIVE GOOGLE TRENDS (INDIA)\n")

    for entry in feed.entries[:10]:
        keyword = entry.title

        # Extract real search volume from ht:approx_traffic tag
        search_volume_raw = entry.get("ht_approx_traffic", "0")
        # Clean it — Google sends values like "1K+", "200+", "10K+"
        search_volume_str = search_volume_raw.replace("K+", "000").replace("+", "").strip()
        try:
            search_volume = int(search_volume_str)
        except:
            search_volume = 100  # fallback only

        event = {
            "timestamp": datetime.utcnow().isoformat(),
            "keyword": keyword,
            "search_volume": search_volume,  # ← real value now
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

python google_trends_rss_live.py

🔥 LIVE GOOGLE TRENDS (INDIA)

Sent: weather new delhi | Volume: 100
Sent: ज्ञानेश कुमार | Volume: 2000
Sent: वेदर | Volume: 100
Sent: war update | Volume: 2000
Sent: తేనీ | Volume: 200
Sent: cboe volatility index | Volume: 100
Sent: akshay kumar bhoot bangla trailer | Volume: 5000
```




Trend Dataset
Data updated 4/1/26
Search
Pages
Page 1
Page 2
File
Export
Share
Explore
Subscribe
Set alert
Monitor
Edit
Copilot
5.00K
Max of peak_search_volume
853.85
Average of avg_search_volume
11
Count of trend_score
Average of avg_search_volume by window_end and keyword
keyword accuweather carlos alcaraz jammu weather today's weather
5K
0K
1:15 AM
window_end
Filters
Search
There aren't any filters to display.
Sum of avg_search_volume by keyword and trend_status
trend_status Emerging Normal
weather today
today's weather
weather new delhi
weather jammu
accuweather
carlos alcaraz
jammu weather
keyword
0K
5K
Sum of avg_search_volume
keyword | Sum of trend_score | Sum of avg_search_volume | Sum of peak_search_volume | trend_status | window_end
weather today | 9 | 5,000.00 | 5,000.00 | Emerging | 04/07/26 01:15:
today's weather | 9 | 500.00 | 500.00 | Emerging | 04/07/26 01:15:
weather new delhi | 5 | 500.00 | 500.00 | Emerging | 04/07/26 01:15:
weather jammu | 9 | 200.00 | 200.00 | Emerging | 04/07/26 01:15:
accuweather | 9 | 100.00 | 100.00 | Normal | 04/07/26 01:15:
carlos alcaraz | 9 | 100.00 | 100.00 | Normal | 04/07/26 01:15:
jammu weather | 4 | 100.00 | 100.00 | Normal | 04/07/26 01:15:
Total | 90 | 11,100.00 | 11,100.00
114%

RESULT

Results of the Cloud-Based Financial Transaction Analytics Project

The implementation of the **"Early Trend Detection using Cloud-Based Analytics & Power BI"** system produced meaningful real-time results that demonstrate the effectiveness of cloud-based streaming analytics. The system successfully captured live trending topics from **Google Trends** and processed them through **Microsoft Azure** cloud services.

The processed output was visualized using **Power BI** dashboards, which provided continuous live visibility into trend movement. The dashboard successfully displayed multiple trend keywords simultaneously. The line chart clearly represented keyword-wise trend movement over time.

Each incoming event updated the dashboard dynamically, enabling continuous monitoring of rising topics. This real-time visualization capability is one of the major outcomes of the project.

The dashboard output proves that the system can continuously monitor live trend fluctuations without manual intervention.

Trend Metric Analysis

The system successfully generated multiple real-time metrics for trend evaluation.

The following key output metrics were obtained:

1. Mention Count

This metric represents the number of times a trend topic appears within the configured time window. It helps identify the frequency of trend occurrence.

2. Average Engagement

This metric represents the average engagement score assigned to a trend topic. It is used to measure the relative intensity of a trend.

3. Peak Engagement

This metric captures the highest engagement value observed for a particular trend. It helps identify sudden spikes.

RESULT

4. Trend Status

The system classifies trends into the following categories:

- Viral
- Emerging
- Normal

This classification is based on predefined threshold logic.

The results clearly show that the system is capable of distinguishing between normal activity and rapidly rising topics.

Performance and Observations

The system performance was observed during live testing sessions.

The following key observations were made:

1. Live Trend Detection

The system successfully detected live trending keywords fetched from Google Trends.

2. Cloud Processing Efficiency

Azure Event Hub and Stream Analytics processed the incoming events efficiently without noticeable delay.

3. Dashboard Accuracy

The Power BI dashboard accurately reflected incoming trend changes.

4. Spike Visibility

Sudden spikes in engagement were clearly visible through the line chart. This confirms the system's ability to support early trend identification.

Overall Result

The results clearly demonstrate that the proposed system successfully achieved the project objectives. The system provides a scalable and reliable framework for real-time trend monitoring and business intelligence.

CONCLUSION

The “Early Trend Detection using Cloud-Based Analytics & Power BI” project successfully demonstrates the practical application of cloud computing and real-time business intelligence systems in detecting emerging trends from live internet search activity. The project successfully achieved all its primary objectives. A real-time trend monitoring pipeline was designed and implemented using Microsoft Azure cloud services and Power BI.

The system continuously collected live trending topics from Google Trends, streamed them into Azure Event Hub, processed them through Azure Stream Analytics, and visualized the results using interactive Power BI dashboards. The implementation clearly demonstrates how cloud-based streaming architecture can be used for continuous event monitoring and real-time analytics.

The project successfully calculated multiple trend evaluation metrics, including:

- mention count
- average engagement
- peak engagement
- trend classification

The use of threshold-based trend status classification enabled the system to identify rising and viral topics at an early stage. This directly supports the main objective of early trend detection. The Power BI dashboard provided clear visual representation of trend movement, engagement spikes, and keyword-wise live activity. This significantly improved the interpretability of the results. One of the major strengths of the project is its scalability. Since the architecture is built completely on cloud services, the system can be extended to handle significantly larger data streams.

The project also provided practical exposure to:

- cloud computing
- real-time event streaming
- dashboard visualization
- live analytics systems
- business intelligence tools

Overall, the project successfully demonstrates how cloud-based analytics platforms can be used for early trend intelligence and real-time decision support.

This work can serve as a strong foundation for future enhancements
such as:

- X / social media trend integration
- news feed analytics
- sentiment analysis
- predictive trend forecasting
- machine learning-based anomaly detection

The outcomes of this project highlight its relevance in real-world applications such as media monitoring, sports analytics, marketing intelligence, and live event tracking.

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